

# MEVIN JOSE

mjenius1357@gmail.com — +91 9902932228 — Bangalore, India  
linkedin.com/in/mevin-jose — mjenius.github.io/Portfolio — github.com/mjenius

## Summary

AI/ML undergraduate specializing in agentic systems, evaluation-driven ML pipelines, and explainable RL. Built a multi-agent SQL system with 95-100% accuracy on a 100-query benchmark with schema-grounded validation, a PPO traffic optimization system improving emergency transit time by 10.6%, and real-time CV pipelines sustaining <100ms latency. Focused on system design, validation rigor, and production-ready ML workflows.

## Skills

**ML & AI (Core):** PyTorch, Scikit-learn, LightGBM, SHAP, OpenCV, NLP

**Agentic & LLM Systems:** RAG, FAISS, Sentence Transformers, Prompt Engineering, Multi-Agent Systems, Llama 3, Groq/Gemini APIs, LangChain, LangGraph, Hugging Face Transformers, Fine-tuning (PEFT, LORA)

**Systems & MLOps:** FastAPI, Docker, CI/CD, GitHub Actions, Drift Detection, Model Monitoring, AWS (EC2, S3)

**Languages:** Python, SQL, JavaScript, TypeScript, Java

**Databases & Frameworks:** PostgreSQL, MongoDB, SQLite, Redis, Pinecone (Vector DBs), Streamlit, Flask, Pandas, NumPy, SciPy, Socket.IO

## Experience

### Superhhero Learning

Bangalore, India

Web Developer Intern

Jun – Sep 2025

- Developed and deployed 8+ frontend pages using the MERN stack serving 100+ active users; optimized MongoDB indexing and query plans to reduce connected API latency by 25%.
- Improved application load speed by 10% and cut engineer iteration time by 25% by architecting automated performance observability pipelines and asset compression workflows.

## Projects

### AI Data Analyst Agent

*Python, FastAPI, FAISS, Sentence Transformers, Groq/Gemini APIs, RAG, Pydantic*

- Achieved 95-100% natural-language-to-SQL accuracy on a 100-query benchmark spanning 8 e-commerce domains, validated against schema-grounded ground truth (expected tables and metrics).
- Architected planner-executor-evaluator multi-agent system with FAISS-based schema retrieval, SQL validation, and multi-model fallback (Groq → Gemini).
- Integrated Dockerized execution sandboxes to isolate dynamically generated SQL scripts and run automated schema validation checks safely.

### Adaptive Traffic Signal Control (PPO)

*PPO, SUMO, Stable-Baselines3, SHAP*

- Reduced emergency vehicle transit time by 10.6% vs fixed baseline using PPO-based reinforcement learning.
- Applied SHAP-based interpretability over 43D state space for policy reliability in safety-critical scenarios.

### Football Recruitment & Scouting System

*Scikit-learn, SHAP, Llama 3 (Ollama), FastAPI, Docker, Streamlit*

- Built clustering and similarity search pipeline over 3,364 players with valuation model ( $R^2 = 0.82$ ), reducing manual scouting time by ~70%.
- Deployed Dockerized FastAPI inference with reproducible pipelines and Streamlit dashboard for stakeholder-facing analytics.

### AEGIS – Tamper-Resistant Surveillance System

*OpenCV, HMAC, Socket.IO, Flask, FFmpeg*

- Built real-time CV pipeline detecting tampering (blur, glare, replay attacks) with <100ms latency on 720p streams.
- Secured live video pipelines against active stream tampering by implementing HMAC-based cryptographic watermark validation coupled with a real-time Socket.IO alerting dashboard.

### Demand Forecasting ML System

*LightGBM, FastAPI, Pandas, SciPy, SQLite, Parquet*

- Cut forecasting error by 25% MAPE across 30K+ SKUs (~2M+ records) via time-aware cross-validation.
- Mitigated production model degradation and concept drift over 30K+ SKUs by engineering an automated rolling-window data monitoring and reproducible retraining pipeline.
- Established baseline thresholds for population stability index (PSI) to trigger automated alerts upon data drift detection.

## Education

### PES University

Bangalore, India

B.Tech, Computer Science (AI & Machine Learning)

Aug 2023 – May 2027

CGPA: 7.85

**Coursework:** Explainable AI, LLMs & Applications (RAG), Advanced Foundations of ML (Deep Learning, Reinforcement Learning), Machine Learning, Optimizations & Metaheuristics, Design and Analysis of Algorithms, Python

## Certifications

Google Data Analytics — Meta GenAI — Microsoft Computer Vision — Kaggle ML & DL